

Andy Ren

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EXPERIENCE

QE Home | Quilts ETC

Software Engineer (Forward Deployed)

Burnaby, BC

Aug 2025 – Present

- Own the Retail Pro Prism integration end to end: designed and built Django/Celery/RabbitMQ middleware that syncs customer, inventory, document, and transfer-order data between Prism's Oracle backend and internal SQL Server systems for 82 retail stores, running ~550 sync jobs and processing ~50K records/day (10M+ records to date).
- Migrated ~980K customer records into Prism by mass-generating ~1M JSON import files with automated 7-Zip batching for PrismJSONImporter, replacing a manual per-record import process.
- Designed and shipped the purchase order planner used by 12 PO managers to plan ~20 POs/month against a 14K-SKU catalog, including an ETA calendar and FOB shipping-cost logic; migrated 1,388 legacy Planner POs (12K line items) into it and retired the spreadsheet workflow.
- Reverse-engineered Prism's sparsely documented REST API (seated vs. non-seated session auth, custom API user provisioning) and Oracle schema to build transfer-order and inventory integrations the vendor does not support, including a nightly auto-ship job that pushes ~2,000 store transfer orders/month.
- Wrote stored procedures and bulk data operations against production Oracle and SQL Server databases, including forensics and cleanup on a 1.17M-row store-level inventory min/max table (115K audited row changes).
- Sole engineer embedded with operations staff: gather requirements, ship, and support internal tooling for 250+ users (~200 monthly active) without a dedicated PM or QA function.

Software Developer Co-op (Full-stack)

May 2023 – Dec 2023

- Built the internal web app's authentication system end to end (Django backend, React frontend, self-serve password recovery), eliminating manual password resets by IT for store staff.
- Built the original purchase order creation workflow and PO ETA calendar that the current planner grew out of.

PROJECTS

Ante — iOS app where screen time is a currency earned by exercise

2026 – Present

React Native (Expo), TypeScript, iOS Screen Time APIs (Family Controls, DeviceActivity, ManagedSettings), HealthKit, Jest

- Built an entirely on-device iOS app (no backend, account, or analytics) where distracting apps stay shielded until a verified HealthKit workout or gym check-in earns minutes, spent in 15-minute blocks; the app and three Screen Time extensions share an append-only ledger through an App Group container.
- Designed the unlock economy around iOS's declarative enforcement limits (no app code runs when a blocked app opens; usage is observable only at pre-armed thresholds; events can fire late or drop) using prepaid tranches, cumulative-minute event names for reconciliation, and guard thresholds so a dropped event leaks at most one block.
- Kept ledger, tranche planning, reconciliation, DST-safe day boundaries, streaks, and health-to-minutes conversion in pure, Jest-tested TypeScript modules; deterministic transaction IDs make double taps and racing syncs idempotent. Verified on device that shields survive force-quit; Family Controls distribution entitlement granted (TestFlight-ready).

TesMiles — CRA-compliant mileage logger for Tesla drivers

2026 – Present

React Native (Expo), Django REST Framework, Celery, MQTT, Claude API

- Ingests Tesla Fleet Telemetry over MQTT to log trips automatically and generate audit-ready CRA mileage reports for self-employed drivers; receipt capture parsed with the Claude API. [\[App Store link / N users\]](#)

Spots Map — full-stack location-sharing web app

Feb 2025 – Mar 2025

- React, Node.js/Express, and MongoDB app for marking and sharing places on an interactive Mapbox map, with JWT/bcrypt authentication.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL (T-SQL, Oracle SQL), Java, Kotlin

Backend & Data: Django, Django REST Framework, Celery, RabbitMQ, Node.js/Express, SQL Server, Oracle, MongoDB, REST API integration, ETL & data migration

Frontend & Mobile: React, React Native (Expo), Tailwind CSS, Bootstrap

Tools: Git, Linux, MQTT, Firebase, [\[Docker, GitHub Actions, Azure/AWS — add once real\]](#)

EDUCATION

Simon Fraser University — BSc, Computer Science

May 2025 | Burnaby, BC